

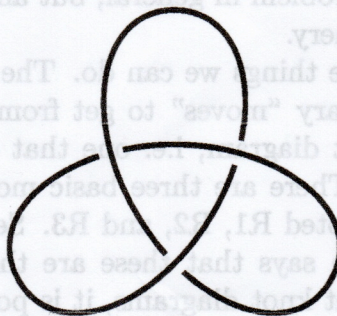
## WORKSHEET 23

### A Taste of Knot Theory

A knot, in mathematics, is a way of taking a string, placing it into three-dimensional space  $\mathbb{R}^3$ , and then gluing the edges together. Because our paper is only 2-dimensional, we cannot draw knots fully, but only projections of them onto the plane  $\mathbb{R}^2$ . But when we project them, we may have two (or more) points on the knot projecting to the same point on the plane. We will always consider projections such that only at most two points ever get projected to the same point on the plane. Thus we need only indicate which point is higher and which is lower. We indicate this by making a gap in the lower strand. For instance, in Figure 1, the red strand is above the blue strand. See Figure 2 for a full knot with all the crossings indicated in this way.



**Figure 1.** The red strand is above the blue strand.



**Figure 2.** A trefoil knot.

When we have two knots, or more precisely knot projections, we would like to know whether it is possible to deform the string continuously to get from one to the other. One question in particular is to determine when we can unknot a knot: can we deform it so that it becomes a circle (or an *unknot* in knot theory speak)? There exist

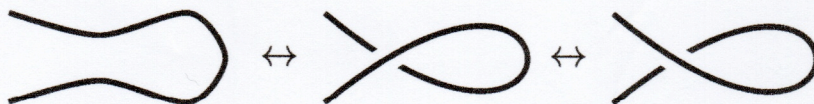


Figure 3. Reidemeister 1.

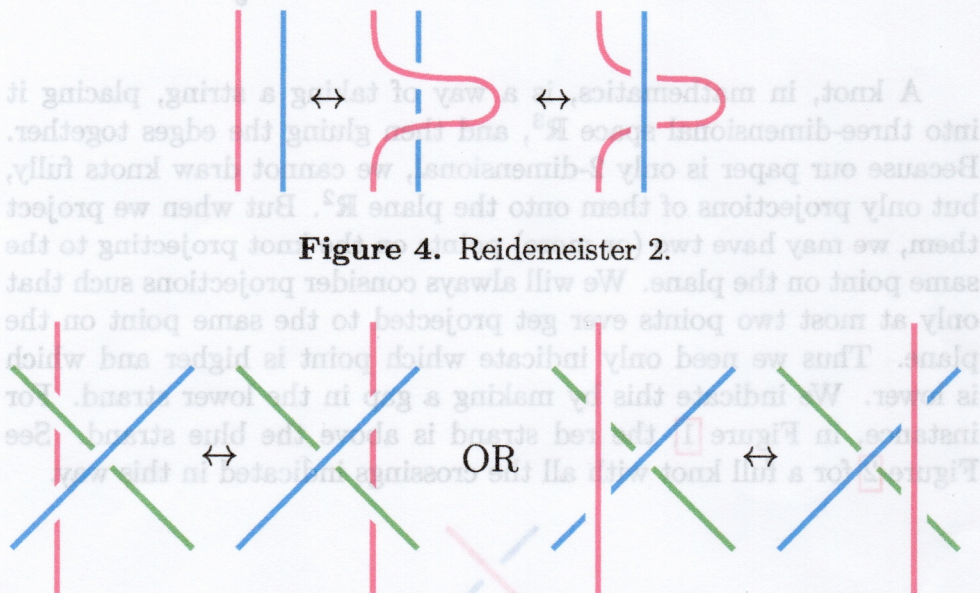


Figure 4. Reidemeister 2.

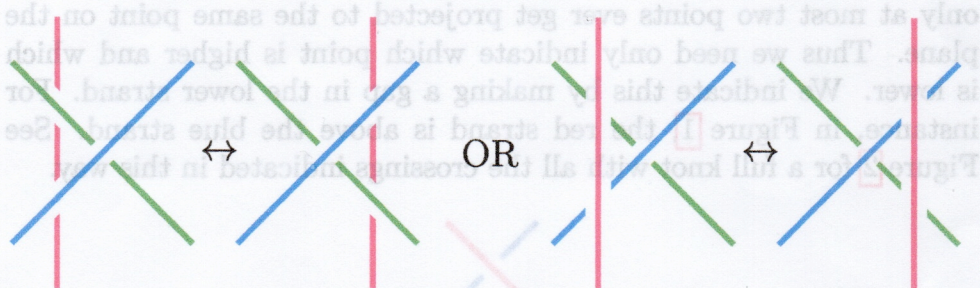


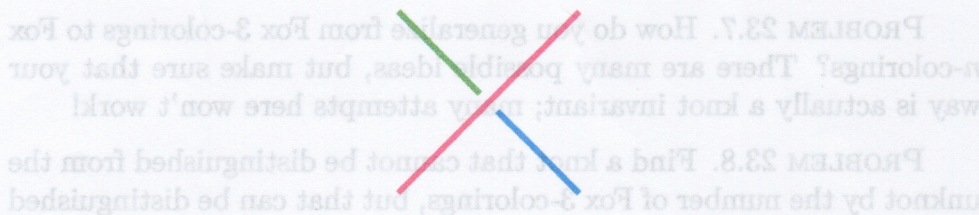
Figure 5. Reidemeister 3.

methods to solve this problem in general, but all of them involve some rather advanced machinery.

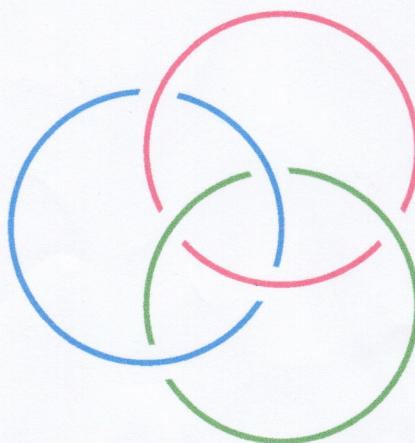
Still, there are some things we can do. The first thing to do is to work out some elementary “moves” to get from one knot diagram to another equivalent knot diagram, i.e. one that can be obtained from deforming the string. There are three basic moves, called the *Reidemeister moves* and denoted  $R_1$ ,  $R_2$ , and  $R_3$ . See Figures 3, 4, and 5. Reidemeister’s Theorem says that these are the only moves needed: given any two equivalent knot diagrams, it is possible to get from one to the other via a sequence of Reidemeister moves.

Let us now use Reidemeister moves to construct some invariants of knots, i.e. functions of knot projections such that two equivalent knots have the same value under the function.

**PROBLEM 23.1.** Suppose that  $I$  is a function on knot projections, say to  $\mathbb{R}$ , i.e. if  $K$  is any knot projection, then  $I(K)$  is a real number. Show that if whenever  $K_1$  and  $K_2$  are related to each other by a Reidemeister move we have  $I(K_1) = I(K_2)$ , then  $I$  is a knot invariant.



**Figure 6.** The three strands emanating from a crossing, marked in different colors.



**Figure 7.** Borromean rings.

One such function is the number of *Fox 3-colorings*. Note that at every crossing in a knot projection, there are three strands emanating from it: the overcrossing strand, and the two broken pieces of the undercrossing strand, as shown in Figure 6. A Fox 3-coloring is a way of coloring all the strands in a knot diagram in such a way that at each crossing, either all three strands are the same color, or else all three strands are different colors.

**PROBLEM 23.2.** Show that the number of Fox 3-colorings is a knot invariant.

**PROBLEM 23.3.** Show that the trefoil knot is not equivalent to the unknot.

**PROBLEM 23.4.** Show that the Borromean rings, shown in Figure 7, cannot be unlinked, even though any two of them are unlinked.

**PROBLEM 23.5.** Show that the number of Fox 3-colorings of a knot is always divisible by 3.

**PROBLEM 23.6.** Show that the number of Fox 3-colorings of a knot is always a power of 3.

PROBLEM 23.7. How do you generalize from Fox 3-colorings to Fox  $n$ -colorings? There are many possible ideas, but make sure that your way is actually a knot invariant; many attempts here won't work!

PROBLEM 23.8. Find a knot that cannot be distinguished from the unknot by the number of Fox 3-colorings, but that can be distinguished by Fox  $n$  colorings for some  $n > 3$ .

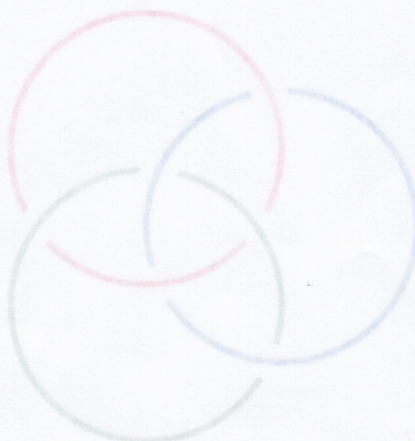


Figure 7. Borromean rings.

One such function is the number of Fox 3-colorings. Note that at every crossing in a knot projection, there are three strands emanating from it: the overcrossing strand, and the two broken pieces of the undercrossing strand, as shown in Figure 6. A Fox 3-coloring is a way of coloring all the strands in a knot diagram in such a way that at each crossing, either all three strands are the same color, or else all three strands are different colors.

PROBLEM 23.2. Show that the number of Fox 3-colorings is a knot invariant.

PROBLEM 23.3. Show that the trefoil knot is not equivalent to the unknot.

PROBLEM 23.4. Show that the Borromean rings, shown in Figure 7, cannot be unlinked, even though any two of them are unlinked.

PROBLEM 23.5. Show that the number of Fox 3-colorings of a knot is always divisible by 3.

PROBLEM 23.6. Show that the number of Fox 3-colorings of a knot is always a power of 3.

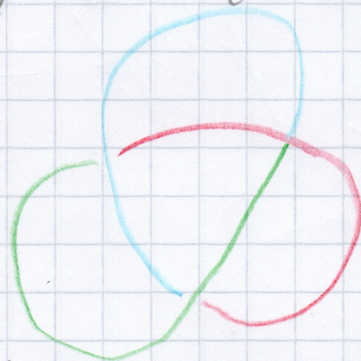
# A Taste of Knot Theory

Alexander Friesen 4/14-19/26

\*Reidemeister

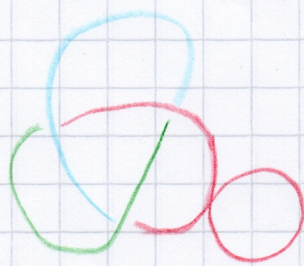
23.1: Because "it is possible to get from one [knot] to the other via a sequence of Reidemeister moves", and because  $I$  is injective (two knots will output the same real number if they are equal themselves),  $I$  is a knot invariant for any 2 equal knots (like  $k_1, k_2$ ).

23.2:

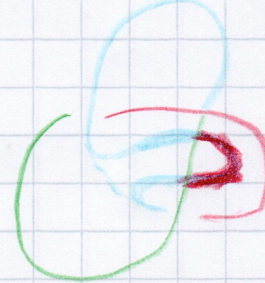


Here's a trefoil colored in 1 of its 9 Fox-3 ways.

What happens if a Reidemeister move is taken?

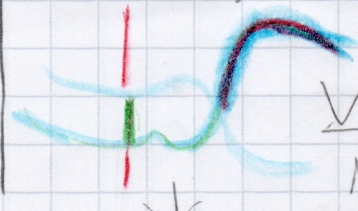


Same but



Valid once the

(Impossible without a few Reidemeister moves first, so just using a subsection)



Valid

No

Color

Change

1 extra intersection. blue 2 strand is red.

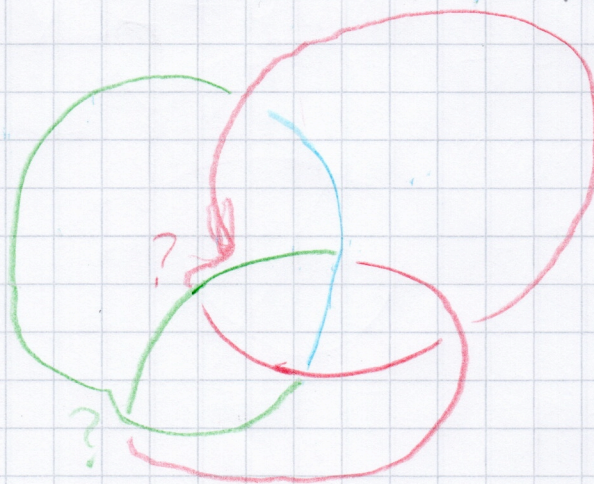
All  $RM^*$  moves keep a Fox-3 coloring. (so an invariant)

23.3: The trefoil knot does have a Fox-3 coloring - 9 of them, too! But an unknot only has three:



Because of 23.2, an unknot and trefoil are not under the same invariant and are therefore  $\neq$ .

23.4:



It turns out that a Borromean ring triple cannot be Fox-3 colored except for via the 3 monochromatic ways. But 3 unlinked rings can each be colored in 3 ways - leading to  $3^3 = 27$  ways to fox-3 color! Not equal knots...

23.5: Any particular Fox 3 coloring can be made into 2 other colorings by swapping every color to the 2 other unused ones. Any fox-3 coloring is in a group of 3 colorings which is part of the set of all possible knot colorings, so that set must be composed of only the size 3 subsets, and therefore is a multiple of 3.

23.6: The most trivial knot has 3 fox-3 colorings, the unknot. Every new non-monochromatic coloring can be both rotated and flipped to create more colorings, ultimately with each rotation and flip combination contributing to multiply all current ways by 3, thus  $3^n$ .

\* This system guarantees unique intersections for each coloring.

23.7: Here is an honest attempt,

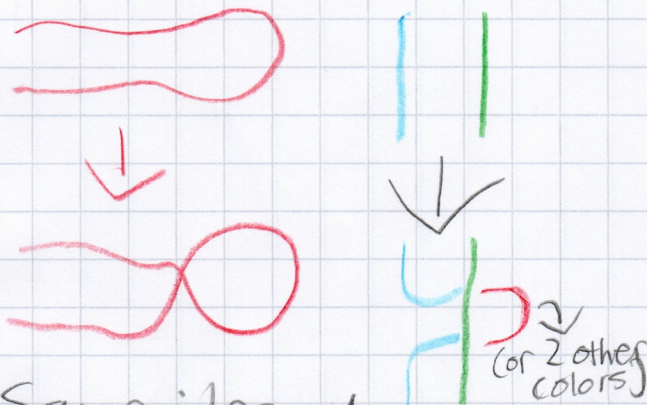
I'm not sure if this is conventional:

Again, 3 colors per intersection,

same format of all same/different.

However, the list of colors to choose from is now size  $n$ .

Reidemister moves proof:



Same idea. Again, same.

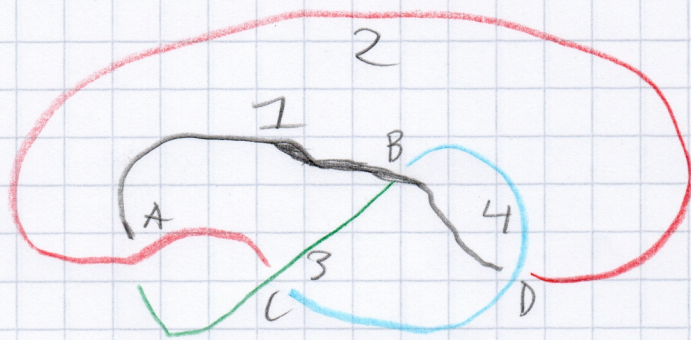
It turns out that

uh oh, this creates more options and isn't invariant!

each intersection is either all same OR averaged (If each color was a number from  $0$  to  $n-1$ , the top is the avg of the 2 bottom.) A bit different,\* but same concept - for 3 the averages are either all same or all different.

\* $\equiv$ , not  $\equiv$ ! <sup>‡</sup>This is the avg function fixed for 5 colors.

23.8: A good example here is the figure 8/Listing's knot, commonly used on ships. Start with fox 3:



Currently, the black line cannot be red or blue. But if  $n=5$  where black is now a color, the diagram is valid as its own knot. Assuming (black, red, green, blue) = (1, 2, 3, 4), applying  $2 \cdot \text{over} \stackrel{*}{=} \text{under} + \text{under} \stackrel{\ddagger}{=} (\text{mod } 5)$  to each intersection gets:

- A:  $2 \cdot 2 = 3 + 1 = 4$ , checks out!
- B:  $2 \cdot 1 = 3 + 4 = 7 \equiv 2 \pmod{5}$ , checks out!
- C:  $3 \cdot 2 = 1 + 4 = 5 \equiv 0 \pmod{5}$ , checks out!
- D:  $4 \cdot 2 = 1 + 2 = 3 \pmod{5}$ , checks out!

All intersections are valid, so the knot fox 5 coloring is too!